

PDF ANONIMIZADO

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n, breeding and vaccination records, of the animal under study, are reported in Table 1.

No of sample	Background information					
	Age	Breed	Sampling site	Location	Reproduction record	Vaccination (annually)
1	32	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
2	9	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
3	13	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
4	7	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
5	12	Native Greek	A	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus
6	20	Irish sport horse	A	Attika Prefecture	AI* once - Did not conceive	Influenza-Tetanus/ Herpesvirus
7	10	Bulgarian sport horse	A	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus
8	18	KWPN (Holland)	A	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus/ Herpesvirus
9	6	Native Greek	A	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus
10	10	British sport horse	A	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus/ Herpesvirus
11	4	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
12	7	Native Greek	A	Attika Prefecture	AI once – 1 successful** gestation	Never vaccinated
13	20	Native Greek	A	Attika Prefecture	AI once - Did not conceive	Influenza-Tetanus
14	11	Bulgarian horse	A	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus
15	11	Native Greek	A	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus
16	25	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
17	15	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
18	16	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
19	16	Native Greek	A	Attika Prefecture	Never mated or inseminated	Never vaccinated
20	26	Arabian	B	Boeotia Prefecture	Never mated or inseminated	Influenza-Tetanus
21	12	Arabian	B	Boeotia Prefecture	AI once/gestation – 5 successful gestations	Influenza-Tetanus
22	5	Arabian	B	Boeotia Prefecture	AI once - Did not conceive	Influenza-Tetanus
23	16	Arabian	B	Boeotia Prefecture	Never mated or inseminated	Influenza-Tetanus
24	24	Arabian	B	Boeotia Prefecture	Never mated or inseminated	Influenza-Tetanus
25	3	Arabian	B	Boeotia Prefecture	Never mated or inseminated	Influenza-Tetanus
26	18	Native Greek	C	Attika Prefecture	Never mated or inseminated	Never vaccinated
27	24	Native Greek	C	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus
28	24	Haflinger	C	Attika Prefecture	Never mated or inseminated	Influenza-Tetanus
29	20	Native Greek	C	Attika Prefecture	AI once/gestation – 2 successful gestations	Influenza-Tetanus
30	12	Native Greek	C	Attika Prefecture	Never mated or inseminated	Never vaccinated

*AI: Artificial insemination; **Gestation that resulted in delivery of a viable foetus

Sampling was conducted aseptically from the inner (cranial) vaginal wall (caudal and ventral to the cervix), using speculum and sterile cotton swabs attached to probes. After sample collection, the swabs were immersed into 2.0 mL Dulbecco's modified Eagle's medium (DMEM) and stored in dry ice, until transportation to the laboratory. There, they were centrifuged (10,000g, 3 min.) and the supernatants (1 mL) were divided into two aliquots that were stored at -20°C (test sample) and -80°C (backup sample).

Results and discussion

were retrieved from 28 of the 30 samples (93.5%), which resulted to the presumptive bacterial species (Supplementary Table S1)¹ belonging to 87 genera (Supplementary Table S2). The two most frequently detected bacterial genera were *Staphylococcus* spp. 89% (25 of 28) and *Acinetobacter* spp. 54% (15 of 28). The following genera were detected in at least 25% of the test samples: *Staphylococcus* (89%), *Acinetobacter* (54%), *Klebsiella* (43%), *Pseudomonas* (36%), *Psychrobacter* (36%), *Bacillus* (32%), *Enterococcus* (25%), *Providencia* (25%) and *Sphingomonas* (25%) (Table 2).

Table 2. The bacterial genera detected in at least 25% of the test samples.

Bacterial genus	Positive samples (%)
<i>Acinetobacter</i>	54%
<i>Bacillus</i>	32%
<i>Enterococcus</i>	25%
<i>Klebsiella</i>	43%
<i>Providencia</i>	25%
<i>Pseudomonas</i>	36%
<i>Psychrobacter</i>	36%
<i>Sphingomonas</i>	25%
<i>Staphylococcus</i>	89%

The relevant findings with regards to the detected bacterial species are recorded in Table 3.

Table 3. The bacterial species detected in at least 25% of the test samples.

Bacterial species	Positive samples (%)
<i>Acinetobacter bouvetii</i>	29%
<i>Acinetobacter equi</i>	32%
<i>Acinetobacter lwoffii</i>	25%
<i>Acinetobacter variabilis</i>	25%
<i>Klebsiella oxytoca</i>	46%
<i>Psychrobacter faecalis</i>	25%
<i>Psychrobacter pulmonis</i>	25%
<i>Staphylococcus cohnii</i>	25%
<i>Staphylococcus equorum</i>	68%
<i>Staphylococcus hominis</i>	25%
<i>Staphylococcus saprophyticus</i>	46%
<i>Staphylococcus succinus</i>	29%

Regarding the number of NGS reads per bacterial genus, scores ranged between 7 and 326,500. The highest scores were recorded for *Enterobacter* (326,500), *Porphyromonas* (102,705), *Frigoribacterium* (54,957), and *Gracilibacter* (50,674), whereas the lowest for *Bacillus* (7), *Ruminococcus* (10), *Staphylococcus* (12), *Acinetobacter* (13), and *Buttiauxella* (16) (Supplementary Figure S1).

For the bio-informatics analysis, the individuals of the study population were assigned to groups based on breed, breeding record, and vaccination, to identify similarity or discriminating traits, with regards to the detected bacterial genera (Supplementary Figure S2, S3, S4). Factors potentially influencing the study parameter, such as the stage of the estrus cycle, nutrition, housing, and management, for which the test population presented little or no variation, were not included in the analysis.

None of the detected bacterial genera or species that were present in all of the samples, indicating that the vaginal bacterial microflora of the study population was diverse, despite its relevant homogeneity regarding location, nutrition, disease, and reproduction record. However, based on the percentage of detection of bacterial genera in the tested samples (> 25%) and the score of the NGS reads (> 150,000), it can be stated that

knowledge, the investigation of the constitution of the vaginal bacterial microflora of een the subject of only three other studies: Barba et al., (2020), Husso et al., (2020) and 22). A comparative assessment of the results of these studies does not provide grounds for safe assumptions about the consistency of their findings (Table 4).

Table 4. Comparison of the results recorded in studies focused on the investigation of the constitution of the vaginal microflora in healthy mares using NGS, in connection with the bacteria more commonly detected.

Bacterial genus detected	Reference			
	Mataragka et al., 2023	Malaluang et al., 2022	Barba et al., 2020	Husso et al., 2020
<i>Acinetobacter</i>	+			
<i>Akkermansia</i>			+	
<i>Arcanobacterium</i>			+	
<i>Bacillus</i>	+			
<i>Campylobacter</i>			+	+
<i>Corynebacterium</i>			+	+
<i>Enterococcus</i>	+			
<i>Escherichia coli</i>		+		
<i>Fusobacterium</i>			+	
<i>Helcococcus</i>				+
<i>Klebsiella</i>	+			
<i>Porphyromonas</i>			+	+
<i>Providencia</i>	+			
<i>Pseudomonas</i>	+			
<i>Psychrobacter</i>	+			
<i>Sphingomonas</i>	+			
<i>Staphylococcus</i>	+	+ [*]		
<i>Streptococcus</i>		+ ^{**}	+	
uncultured <i>Kiritimatiaellae</i>			+	

^{*}*Staphylococcus capitis*; ^{**}*Streptococcus equisimilis*, *S. thoralensis*, *S. zooepidemicus*

This could be associated with the dynamic interaction between the constitution of the vaginal microbiota and several environmental or physiological parameters, as well as methodological factors (Barba et al., 2020; Uchihashi et al., 2015; Yildirim et al., 2014). In this regard, it is perhaps worth noting that in the study of Malaluang et al., (2022), the bacterial species more commonly detected was *Escherichia coli*, which is consistent with the close proximity of the vagina to the anus. Interestingly, *E. coli* was not among those most commonly detected in the samples of other two NGS studies, which could be associated with the method or the anatomical site of sample collection. As already mentioned, the swabs analyzed in the study presented here were taken from the inner vaginal wall, whereas those of the other NGS studies are reported to be the vestibule-caudal vagina (Barba et al., 2020), the ventral side of the vaginal vestibulum (Husso et al., 2020), and the vaginal fornix of the cranial vagina (Malaluang et al., 2022).

Conclusion

The bio-informatics analysis conducted with regards to breed, produced evidence of a much richer microbiota for the Greek native (60 bacterial genera), compared to the Arabian (37) and other breeds (23). With regards to the Arabian breed, which was included in the study population of Barba et al., (2020), the number of the bacterial genera detected was 16, and 6 of them were also detected in our study: *Peptoniphilus*, *Campylobacter*, *Ruminococcus*, *Streptococcus*, *Helcococcus* and *Porphyromonas* (Table 4). Notably, eight (8) bacterial genera were detected across breeds. With the exception of *Clostridium* spp., these bacteria were the same with those reported above to be more commonly detected in the study population at high NGS score-reads, which is consistent with the certain bacteria being, as already mentioned, the main elements of the microbiota of the test population (Supplementary Figure S2). The relevant results recorded with regards to reproduction record (Supplementary Figure S3) and vaccination (Supplementary Figure S4), indicate that these parameters may also be associated with variations of the

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